Subham Kumar Sahu

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Summary

Computer Engineering undergraduate with research interests in Computational Intelligence, Metaheuristic Optimization, Swarm Intelligence, and Machine Learning. Developed a novel Multi-Objective Penguin Search Optimization Algorithm (MOPeSOA), with a manuscript currently under review at Heliyon. Currently designing a new Flying Fish Optimization Algorithm for single-objective optimization. Experienced in algorithm design, benchmark evaluation, scientific writing, and AI-based problem solving. Seeking Research Associate opportunities in optimization, intelligent systems, and computational research.

Education

C. V. Raman Global University — **Nov 2022 – May 2026**

Bachelor of Technology in Computer Engineering | CGPA: 8.11 — *Bhubaneswar, Odisha*

Research Experience

Multi-Objective Penguin Search Optimization Algorithm (MOPeSOA) — **2025 – Present**

First Author — *CV Raman Global University*

- Developed a novel multi-objective metaheuristic optimization algorithm extending the Penguin Search Optimization Algorithm (PeSOA) to multi-objective optimization problems.
- Introduced Pareto-dominance based archive maintenance, oxygen-adaptive search control, cooperative multi-group exploration, and Differential Evolution based search mechanisms.
- Compared performance against NSGA-III, MOEA/D-DE, MOPSO, MOHHO, MOWOA and MOGWO.

Flying Fish Optimization Algorithm (FFOA) — **2026 – Present**

Researcher — *CV Raman Global University*

- Developing a new single-objective swarm intelligence optimization algorithm inspired by flying fish locomotion behavior
- Investigating adaptive exploration-exploitation balancing mechanisms for complex optimization landscapes.
- Performing benchmark evaluation and comparative analysis against state-of-the-art metaheuristic algorithms.

Publications

- **Subham Kumar Sahu**, Rahul Kumar Gupta, Adyasha Rath, Prabodh Kumar Sahoo, Aswini Kumar Samantaray, Ganapati Panda.

  "Development of a Novel Multi-Objective Penguin Search Optimization Algorithm Using Adaptive and Cooperative Search Strategies."

  Under Review at Heliyon (Elsevier).

Projects

Potato Plant Disease Classification System — **TensorFlow, FastAPI, React.js, Docker**

- Developed an **image classification system** using **TensorFlow** to detect potato plant diseases from leaf images.
- Trained and integrated the model with a **FastAPI backend** for real-time prediction.
- Built a **React.js frontend** enabling users to upload images and view disease detection results instantly.

Technical & Research Skills

Research Areas: Computational Intelligence, Swarm Intelligence, Metaheuristic Optimization, Multi-Objective Optimization, Evolutionary Computing, Machine Learning
Programming: Python, Java, C, SQL
Libraries: NumPy, Pandas, Scikit-Learn, TensorFlow, XGBoost, Matplotlib
Research Skills: Algorithm Design, Benchmark Function Analysis, Scientific Writing, Experimental Design
Tools: Git, GitHub, Docker, Linux, Google Colab

Relevant Coursework

- Artificial Intelligence
- Machine Learning
- Design and Analysis of Algorithms
- Data Structures and Algorithms
- Probability and Statistics
- Database Management Systems